

ONLINE EXAM MONITORING SYSTEM USING AI-BASED MONITORING

An Major Project (Stage-II) Report Submitted to

Jawaharlal Nehru Technological University Hyderabad

*In partial fulfillment of the requirements
for the award of the degree of*

**BACHELOR OF TECHNOLOGY
IN
COMPUTER SCIENCE AND ENGINEERING**

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**DEPARTMENT OF COMPUTER SCIENCE AND ENGINEERING
BHARAT INSTITUTE OF ENGINEERING AND TECHNOLOGY**

(An Autonomous Institutions)

Accredited by NAAC "A" Grade, Accredited by NBA (UG Programmes: CSE, ECE)

Approved by AICTE, Affiliated to JNTUH Hyderabad

Ibrahimpattam -501 510, Hyderabad, Telangana

DECEMBER 2025



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Certificate

*This is to certify that the Major Project(Stage -II) work entitled
"ONLINE EXAM MONITORING SYSTEM USING AI-BASED MONITORING" is
the bonafide work done*

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ACKNOWLEDGEMENT

The satisfaction that accompanies the successful completion of the task would be put incomplete without the mention of the people who made it possible, whose constant guidance and encouragement crown all the efforts with success.

We avail this opportunity to express our deep sense of gratitude and hearty thanks to **Sri CH. Venugopal Reddy**, Chairman of BIET, for providing congenial atmosphere and encouragement.

We would like to thank **Prof. G. Kumaraswamy Rao**, Former Director & O.S. of DLRL Ministry of Defence, Sr. Director R&D, BIET, and **Dr. V Srinivasa Rao**, Dean CSE, for having provided all the facilities and support.

We would like to thank our Department Incharge **Mr. Romy Sinha** for encouragement at various levels of our Project.

We are thankful to our Project Coordinator **Dr. Rama Prakasha Reddy CH.** Assistant Professor, Computer Science and Engineering for his support and cooperation throughout the process of this project.

We are thankful to our guide **Mrs. Sukanya Kulkarni** Assistant Professor, Computer Science and Engineering for her sustained inspiring Guidance and cooperation throughout the process of this project. Her wise counsel and suggestions were invaluable.

We express our deep sense of gratitude and thanks to all the Teaching and Non-Teaching Staff of our college who stood with us during the project and helped us to make it a successful venture.

We place highest regards to our Parent, our Friends and Well-wishers who helped a lot in making the report of this project

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DECLARATION

We hereby declare that this **Major Project (Stage – II)** is titled **ONLINE EXAM MONITORING SYSTEM USING AI-BASED MONITORING** is a genuine Major Project (Stage -II) work carried out by us, in **B.Tech (Computer Science and Engineering)** degree course of **Jawaharlal Nehru Technology University Hyderabad, Hyderabad** and has not been submitted to any other course or university for the award of my degree by us.

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ABSTRACT

In the age of digital education, conducting secure and fair online examinations has become a significant challenge. Traditional online tests lack robust monitoring, making them vulnerable to various forms of academic dishonesty. To address this, we have developed a comprehensive **Online Exam Proctoring System**, a full-stack web application designed to ensure the integrity of remote assessments.

This project, titled "Proctor Vision Hub," leverages modern web technologies to create a secure environment for students to take exams from any location. The system monitors candidates in real-time using their webcam and microphone, tracks on-screen activity to prevent tab-switching, and employs AI-powered face detection to ensure the authorized student is present throughout the exam duration.

The platform is built on the MERN stack (MongoDB, Express.js, React.js, Node.js), utilizing WebRTC for live video streaming and WebSockets for real-time communication between the student, server, and proctor. The system features distinct modules for students, teachers, and administrators, offering functionalities such as exam creation, user management, live proctoring dashboards, and automated result generation. This report details the architecture, workflow, key features, and implementation of the system, demonstrating how full-stack development can be applied to build a scalable and effective solution for maintaining academic integrity in online education.

Keywords: Online Proctoring, Full-Stack Development, MERN Stack, WebRTC, AI, Cheating Detection, Academic Integrity.

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LIST OF ABBREVIATIONS

S NO.	SYMBOL	ABBREVIATION
1.	AI	Artificial Intelligence
2.	MERN	MongoDB, Express.js, React.js, Node.js
3.	WebRTC	Web Real-Time Communication
4.	API	Application Programming Interface
5.	UI/UX	User Interface / User Experience
6.	JWT	JSON Web Token
7.	DB	Database
8.	CRUD	Create, Read, Update, Delete
9.	GUI	Graphical User Interface
10.	HTTP	Hypertext Transfer Protocol
11.	IDE	Integrated Development Environment

1. INTRODUCTION

1.1 OVERVIEW OF ONLINE PROCTORING

The proliferation of digital technologies has catalyzed a paradigm shift in the global education sector, with e-learning and remote assessments moving from a niche convenience to a mainstream necessity. While this transition offers unprecedented flexibility and accessibility, it simultaneously exposes a critical vulnerability: the integrity of the academic evaluation process. In an unmonitored remote environment, traditional examinations are susceptible to a wide range of academic misconduct, thereby devaluing the credibility of certifications and degrees. An Online Exam Proctoring System is an advanced technological solution designed to bridge this integrity gap. It functions as a digital invigilator, creating a secure and monitored environment for students to undertake examinations remotely. By leveraging a combination of webcam surveillance, screen sharing, browser-lockdown mechanisms, and Artificial Intelligence, these systems perform three critical functions:

1. Identity Verification: Ensuring the person taking the exam is the registered candidate.
2. Continuous Monitoring: Observing the candidate and their environment for any behavior that violates exam regulations.
3. Integrity Assurance: Creating an evidentiary trail of the exam session through recordings and logs, which can be reviewed to resolve disputes.

1.2 PROJECT OBJECTIVES

The primary goal of this project is to develop a robust, scalable, and user-friendly Online Exam Proctoring System. The main objectives are:

1. **To Develop a Secure, Role-Based Platform:** Engineer a multi-tenant web application with distinct, secure access portals for three user roles: Administrators, Teachers (Proctors), and Students, using JWT-based authentication and authorization.
2. **To Implement a Versatile Exam Management Module:** Create a full-featured module for teachers to perform CRUD (Create, Read, Update, Delete) operations on exams, supporting various question formats (e.g., Multiple Choice, Descriptive), and to configure parameters such as duration, scheduling.

3. **To Engineer a Real-Time Proctoring Engine:** Integrate a core monitoring system using WebRTC for high-quality, low-latency video and audio streaming from the student's device. This engine will also capture the student's entire screen to monitor on-screen activity.
4. **To Automate Anomaly Detection:** Implement a series of automated triggers and alerts for common cheating behaviors, including:
 - Navigating away from the exam tab (tab-switching).
 - The presence of multiple faces or no face in the camera's view.
 - Detection of specific audio signatures (e.g., human speech).
 - The student leaving the examination environment.
5. **To Design an Intuitive Proctoring Dashboard:** Build a live dashboard for teachers to monitor multiple students concurrently, with real-time alerts highlighting suspicious events for immediate manual review.
6. **To Automate the Assessment and Reporting Workflow:** Develop a module for the automatic grading of objective questions upon submission, provide a streamlined interface for manual grading of subjective answers, and generate comprehensive analytical reports on student performance and proctoring incidents.

1.3 SYSTEM ARCHITECTURE OVERVIEW

The system is architected using a modern three-tier design, which decouples the presentation, logic, and data layers to enhance scalability, maintainability, and flexibility.

Client-Side (Presentation Layer): A Single Page Application (SPA) built with **React.js**. This layer is responsible for rendering the entire user interface, capturing user inputs, and managing real-time data display. It communicates with the backend via RESTful APIs for data operations and establishes a persistent WebSocket connection for real-time event handling.

Server-Side (Application/Logic Layer): A backend built on the **Node.js** runtime environment with the **Express.js** framework. This layer houses the core business logic, manages user authentication, serves the REST APIs, and operates the WebSocket server for real-time communication and proctoring event processing.

Database Layer: A **MongoDB** NoSQL database serves as the persistence layer. It stores all system data, including user credentials, exam content, student submissions, and detailed, timestamped proctoring logs, in a flexible, JSON-like document format.

2. MOTIVATION

The impetus for this project stems from a confluence of educational, technological, and social factors that highlight a critical need for trustworthy remote assessment solutions.

Preserving Academic Integrity:

At its core, education relies on the principle of fair and honest evaluation. The uncontrolled nature of remote exams threatens this principle, potentially leading to the devaluation of academic credentials and a crisis of confidence in online learning. We are motivated by the fundamental goal of restoring and upholding academic integrity, ensuring that a student's success is a true measure of their knowledge and effort.

Democratizing Secure Assessment Technology:

Access to high-quality education should not be limited by geography, and similarly, access to secure assessment tools should not be limited by an institution's budget. The market for proctoring services is dominated by commercial vendors with high, recurring costs. This creates an inequity where only well-funded institutions can afford to secure their online exams. We are motivated by the desire to democratize this technology by creating a powerful, open-source alternative that any institution can deploy and customize.

Addressing the Challenge of Scalability and Efficiency:

The manual process of invigilating exams is inefficient even in a physical setting. In an online context, where a proctor can effectively monitor only a handful of students at a time, it becomes completely unscalable for large universities or massive open online courses (MOOCs). Our motivation is to engineer a system that leverages automation to dramatically improve efficiency, allowing a single proctor to confidently oversee a large number of students by focusing their attention only on AI-flagged anomalies.

A Synthesis of Pedagogical Needs and Modern Technology:

This project represents a fascinating intersection of pedagogy and advanced software engineering. It challenges us to translate the principles of effective examination supervision into a robust technological framework. We are driven by the technical challenge of building a real-time, high-concurrency, data-intensive application using a modern full-stack approach, incorporating WebRTC for media streaming, WebSockets for instant communication, and the MERN stack for a cohesive development experience.

3. LITERATURE SURVEY

Slusky (2022) emphasizes the growing cybersecurity and privacy challenges of AI-based proctoring tools, highlighting the need for strong encryption, transparent data handling, and ethical AI usage to prevent misuse and bias. Similarly, Kharbat and Abu Daabes (2021) provide insights into the practical adoption of e-proctored exams during the COVID-19 pandemic, identifying both benefits in maintaining exam integrity and drawbacks such as student stress, technical issues, and privacy concerns.[1]

Hosgurmath et al. (2022) present a face recognition system that combines convolutional feature extraction with a Linear Collaborative Discriminant Regression Classifier (LCDRC) to achieve high-accuracy identity verification. The proposed method first uses convolutional layers to extract deep, discriminative facial features that are robust to variations in lighting, pose, and background.[2]

Duhaim, Al-mamory, and Mahdi (2022) investigate various data mining techniques for detecting cheating behaviors in online examinations during the COVID-19 pandemic, when remote assessments became the default mode in education. Their study focuses on analyzing student interaction data, including mouse movements, answer-change patterns, time spent per question, login activity, and navigation behavior, to identify anomalies that may indicate malpractice.[3]

Radwan, Al Abachy, and Al-Araji (2022) present a comprehensive ten-year survey of the various methods developed to detect student cheating during examinations, covering both traditional and technology-enhanced approaches. Their review classifies cheating-detection techniques into categories such as biometric monitoring, behavioral analytics, statistical detection, network monitoring, AI-based proctoring, and device-usage tracking.[4]

Kharbat and Abu Daabes (2021) provide an in-depth understanding of the implementation and impact of e-proctored examinations during the COVID-19 pandemic, focusing on the experiences of both students and educators. Their research reveals that the sudden shift to online learning forced institutions to adopt remote proctoring technologies rapidly, which helped maintain academic integrity and continuity during lockdowns.[5]

Muzaffar et al. (2021) conduct a detailed systematic review of global online examination solutions within e-learning, examining the techniques, tools, architectures, and proctoring mechanisms adopted worldwide. The authors categorize online exam systems into key components such as authentication methods, question-generation techniques, cheating-detection mechanisms, proctoring tools, and exam-delivery platforms.[6]

Nigam et al. (2021) present a comprehensive analysis of the evolution of AI-based proctoring systems, discussing their past developments, current capabilities, and future research directions. The authors highlight that earlier systems primarily relied on manual or semi-automated monitoring, whereas modern solutions integrate advanced computer vision, facial recognition, gaze tracking, and behavior analysis to detect malpractice in real time.[7]

Raman et al. (2021) investigate the adoption of online proctored examinations among university students during the COVID-19 pandemic through the lens of innovation diffusion theory. Their study analyzes how factors such as perceived usefulness, ease of use, trust, and social influence impacted students' willingness to accept and adapt to online proctoring technologies.[8]

Hussein et al. (2020) conducted a detailed evaluation of various online proctoring tools to assess their functionality, effectiveness, and impact on both students and instructors. The study categorizes proctoring systems into live, recorded, and automated AI-based approaches, analyzing each in terms of security, usability, reliability, and ethical considerations. The authors found that while automated AI-based tools efficiently detect cheating behaviors such as face absence, tab switching, and multiple person detection.[9]

Li et al. (2021) propose a visual analytics-based approach to enhance the proctoring of online examinations by providing proctors with intuitive, real-time visual interfaces for monitoring student behavior. Instead of solely relying on automated AI alerts, their system combines computer vision techniques, behavioral feature extraction, and interactive visualization dashboards to help human proctors quickly identify suspicious actions such as abnormal head movement, eye gaze deviation, or the presence of additional persons.[10]

Ahmad et al. (2021) introduce a deep learning-based online proctoring system that integrates multiple AI techniques face recognition, eye-blinking detection, and object detection—to enhance the accuracy

and reliability of remote exam monitoring. Their system uses facial recognition to verify the student's identity throughout the exam and employs eye-blinking analysis to ensure that a real human is present, reducing the risk of spoofing attacks such as photo or video impersonation. [11]

Atoum et al. (2020) present one of the earliest comprehensive frameworks for automated online exam proctoring, combining several computer vision and machine learning components to monitor students continuously during remote assessments. Their system integrates face detection, face recognition, gaze estimation, and 3D head-pose analysis to verify identity and detect suspicious behaviors such as looking away from the screen or leaving the camera view. [12]

Kiyani et al. (2020) present a robust framework for continuous user authentication using keystroke dynamics, aimed at strengthening security in online systems where identity verification must persist beyond initial login. Their approach introduces a Recurrent Confidence Model (RCM) combined with an ensemble learning technique to capture the temporal patterns of user typing behavior, such as key press duration, latency between keystrokes, and rhythm consistency.[13]

Asep and Bandung (2019) propose a framework for continuous user verification in online exam proctoring systems, specifically targeting mobile learning (m-learning) environments. Their design focuses on ensuring that the authenticated student remains the same individual throughout the entire duration of the exam, addressing the limitations of one-time login verification.[14]

Ghizlane, Hicham, and Reda (2019) propose a novel model for automatic and continuous online exam monitoring, designed to enhance the reliability and security of remote assessments. Their model integrates real-time video analysis, biometric verification, and behavioral tracking to ensure that the authenticated student remains present and compliant throughout the examination.[15]

4. OBJECTIVES & FEASIBILITY STUDY

4.1 OBJECTIVES

The project is guided by a clear set of technical and functional objectives, as outlined in detail in Chapter 1. The overarching goal is to deliver a production-ready Online Exam Proctoring System that is secure, scalable, and user-friendly for all stakeholders involved in the academic assessment process.

4.2 FEASIBILITY STUDY

The ensure the backend database and application logic are designed for scalability to accommodate a growing user base and an expanding knowledge base. To Document and Conceptually Evaluate the System: To comprehensively document the system architecture, design choices, implementation details, and functionalities. To define a framework for conceptual evaluation of the system's usability, the effectiveness of the chatbot's recommendations, the utility of the cost estimation tool, and overall user satisfaction through simulated user scenarios and heuristic evaluation, laying groundwork for future pilot testing.

4.2 Feasibility Study

A thorough analysis was conducted to ascertain the project's viability.

The project is highly economically feasible. The total cost of ownership is significantly lower than commercial alternatives.

Development Costs: The primary cost is the development effort. By leveraging the MERN stack and other open-source libraries, we completely avoid any software licensing or purchasing costs.

Operational Costs: Post-deployment, the costs are limited to cloud hosting (e.g., AWS EC2 for the server, S3 for video storage, and MongoDB Atlas for the database). These costs are usage-based and can be scaled, starting from a very low baseline. This contrasts sharply with the per-student, per-exam pricing model of commercial services, which can range from 15-30 per student, making our solution orders of magnitude cheaper for large-scale deployments. The project is technically sound and achievable with current technology.

Technology Stack: The MERN stack is a mature and widely adopted stack with a vast ecosystem of libraries and extensive community support. Node.js is exceptionally well-suited for handling the many concurrent connections required for real-time proctoring.

Core Technologies: WebRTC and WebSockets are now W3C standards and are natively supported by all modern web browsers, eliminating the need for plugins or custom client-side software.

Expertise: The project team possesses the necessary skills in JavaScript, React, Node.js, MongoDB, and networking protocols to successfully execute the implementation. The availability of AI libraries like TensorFlow.js for potential future client-side processing further enhances feasibility.

Social Acceptance: While online proctoring raises valid concerns about student privacy, this system is designed with a "privacy-first" approach. Monitoring is strictly limited to the exam duration, and students must provide explicit, granular consent before their camera, microphone, or screen can be accessed. This transparency is crucial for social acceptance. The system serves a clear societal good by ensuring the value and credibility of education.

Operational Feasibility: The system is designed with user-friendly interfaces for all roles. Teachers can create and manage exams through an intuitive GUI without needing technical expertise. Students undergo a simple onboarding and system check before an exam. The institution would need to provide minimal training and support, making the system operationally viable.

5. PROBLEM STATEMENT

5.1 EXISTING SYSTEMS AND THEIR LIMITATIONS

The shift to remote learning has exposed a critical flaw in the digital education ecosystem: the absence of a secure and scalable method for conducting high-stakes examinations. The current solutions are inadequate and suffer from several key limitations:

1. **Inherent Insecurity:** Generic tools like online survey platforms or Learning Management System (LMS) quizzes offer no invigilation, making them trivially easy to circumvent through collaboration, use of unauthorized resources, or impersonation.
2. **Lack of Scalability:** Ad-hoc solutions, such as using video conferencing software (e.g., Zoom) for manual proctoring, are operationally inefficient. They require a low proctor-to-student ratio, are difficult to manage for large groups, and produce an unmanageable amount of raw video data with no intelligent analysis.
3. **Prohibitive Costs:** Professional-grade commercial proctoring services, while effective, operate on expensive, proprietary platforms. Their pricing models are often a significant financial barrier for public universities, schools, and educational institutions in developing countries.
4. **Fragmented Workflow:** Institutions are often forced to use a disparate set of tools for exam creation, delivery, proctoring, and grading. This fragmented workflow is inefficient, prone to errors, and provides a disjointed experience for both faculty and students.

5.2 PROPOSED SYSTEM

To address these deficiencies, we propose the design and development of a **unified, full-stack Online Exam Proctoring System**. This system will provide an integrated, end-to-end solution that combines a robust exam management platform with an intelligent, automated proctoring engine.

The core of our proposed solution is a real-time monitoring module that leverages WebRTC and WebSockets to securely stream a student's audio, video, and screen activity to a central server and a live proctor dashboard. The system will not just passively record data but will actively analyze it to generate automated alerts for suspicious behaviors like tab-switching, unauthorized audio, and facial anomalies. By integrating these proctoring features directly into the examination platform, our system will provide a seamless, secure, and scalable workflow for creating, conducting, and grading remote assessments, offering a cost-effective and powerful alternative to existing solutions.

6. DESIGN METHODOLOGY

6.1 SYSTEM REQUIREMENTS SPECIFICATIONS

➤ **Administrator:**

FR1: The admin shall be able to create, view, update, and delete Teacher and Student accounts.

FR2: The admin shall be able to view system-wide analytics, such as the total number of exams conducted.

➤ **Teacher:**

FR3: The teacher shall be able to create an exam with a title, instructions, duration, and schedule.

FR4: The teacher shall be able to add, edit, and delete questions of multiple types (MCQ, Descriptive) to an exam.

FR5: The teacher shall be able to view a live dashboard of all students taking an exam.

FR6: The system shall display real-time alerts on the dashboard when a student violates a proctoring rule.

FR7: The teacher shall be able to manually grade descriptive answers and publish final results.

➤ **Student:**

FR8: The student shall be able to register and log in to the system.

FR9: The student shall be able to view a list of their upcoming and completed exams.

FR10: Before starting an exam, the system shall require the student to grant permissions for the camera, microphone, and screen sharing.

FR11: The student shall be able to answer questions and submit the exam within the allotted time.

FR12: The student shall be able to view their graded results after they have been published.

➤ **NFR1 (Security):** All data transmission between the client and server must be encrypted using HTTPS. Passwords must be hashed using a strong algorithm (e.g., bcrypt).

➤ **NFR2 (Performance):** The application pages must load within 3 seconds. The latency for real-time alerts must be under 500ms.

➤ **NFR3 (Scalability):** The system should be architected to handle at least 100 concurrent exam sessions without significant performance degradation.

- **NFR4 (Reliability):** The system should have an uptime of 99.5% and include mechanisms for error handling and logging.
- **NFR5 (Usability):** The user interface must be intuitive and require minimal training for all user roles.

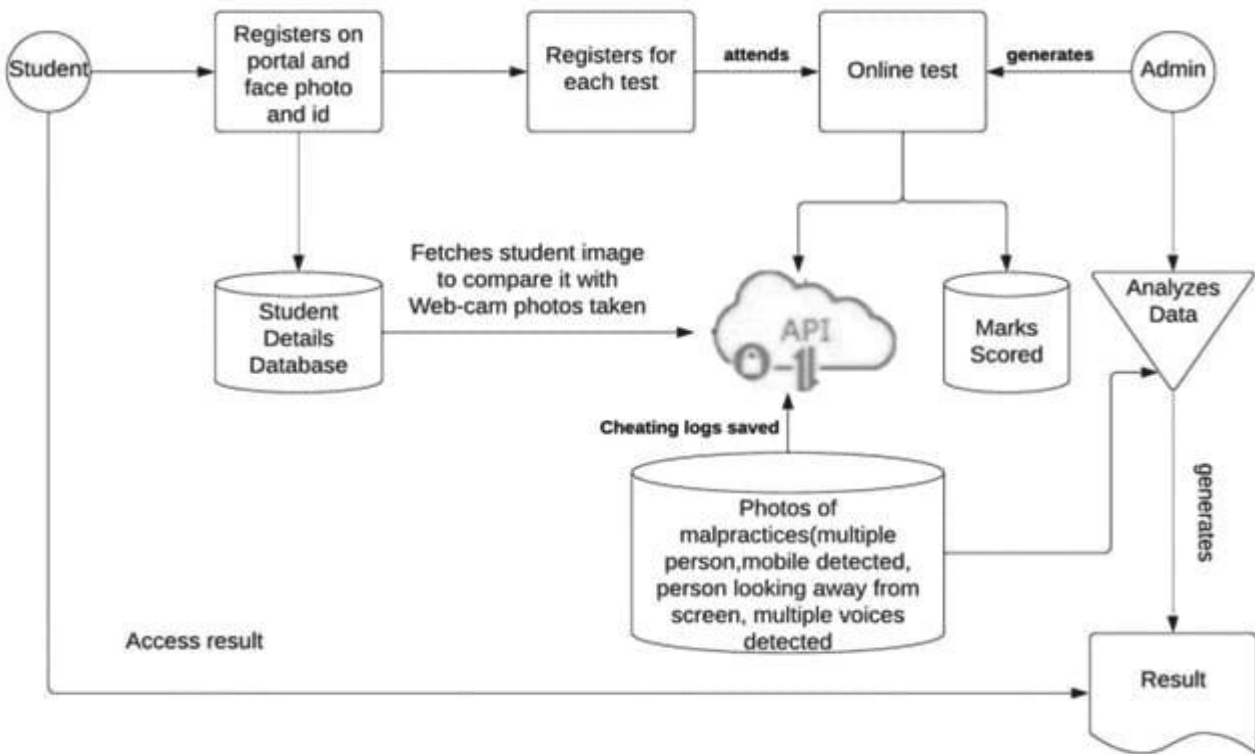


Fig 6.1: System Analysis Architecture Online Exam Monitoring System Using AI-Based Monitoring

The **System Analysis Architecture** diagram of the AI-based Online Exam Monitoring System illustrates the interaction between the major components of the MERN stack architecture and supporting technologies. At the top level, the **User** (student, teacher, or admin) interacts through the **Client Interface**, which is developed using **React.js**. This client layer handles user inputs, renders the interface dynamically, and communicates with the server through **RESTful APIs** and **WebSocket connections** for real-time updates. The **Server**, which is built using **Node.js** and **Express.js**.

6.2 TECHNOLOGY STACK ELABORATION

React.js (Frontend): Chosen for its component-based architecture, which allows for the creation of reusable UI elements (like a question component or a proctoring video feed component). Its virtual DOM ensures efficient updates to the UI, which is critical for a dynamic dashboard displaying live data.

Node.js & Express.js (Backend): Selected for its non-blocking, event-driven I/O model. This makes Node.js exceptionally performant for applications with many concurrent connections, such as the WebSocket connections from every student taking an exam. Express.js provides a minimalist but powerful layer for building the REST APIs.

MongoDB (Database): Its flexible, document-based data model is ideal for this application. An exam can have a varied number of questions of different types, and proctoring logs are unstructured events all of which are easily stored in MongoDB without a rigid schema. Its high write throughput is also advantageous for logging real-time events.

WebRTC & Socket.IO (Real-Time Communication): These two technologies work in tandem. WebRTC handles the heavy lifting of peer-to-peer (or through a TURN server) streaming of high-bandwidth media data (video/audio). Socket.IO acts as the lightweight "signaling" server to coordinate the WebRTC connection setup and is used to transmit low-latency, small-payload event data, such as "user switched tab" alerts, efficiently.

6.3 CORE MODULES DESCRIPTION

1. **Authentication Module:** Uses JWT. Upon successful login, the server generates a signed token containing the user's ID and role. This token is sent to the client and included in the header of all subsequent API requests. The server validates this token on every request to authorize the user and enforce role-based access.
2. **Exam Management Module:** A set of RESTful endpoints (e.g., POST /api/exams, GET /api/exams/:id) that allow teachers to manage the entire lifecycle of an exam. The data is stored in MongoDB in collections like exams and questions.
3. **Proctoring & Monitoring Module:** On the client-side, JavaScript event listeners (window.addEventListener('blur', ...)) detect proctoring violations. When an event is triggered, a JSON payload {eventType: 'TAB_SWITCH', timestamp: Date.now()} is emitted to the Socket.IO server. The server receives this event, creates a new document in the proctoringLogs collection, and then

broadcasts this event to the specific teacher's dashboard room, causing the UI to update in real-time.

4. **Result Management Module:** When an exam is submitted (POST /api/submissions), the backend logic iterates through the submitted answers, compares them against the correct answers stored in the questions collection for MCQs, and calculates an initial score.

6.4 UML DIAGRAMS

6.4.1 USE CASE DIAGRAM:

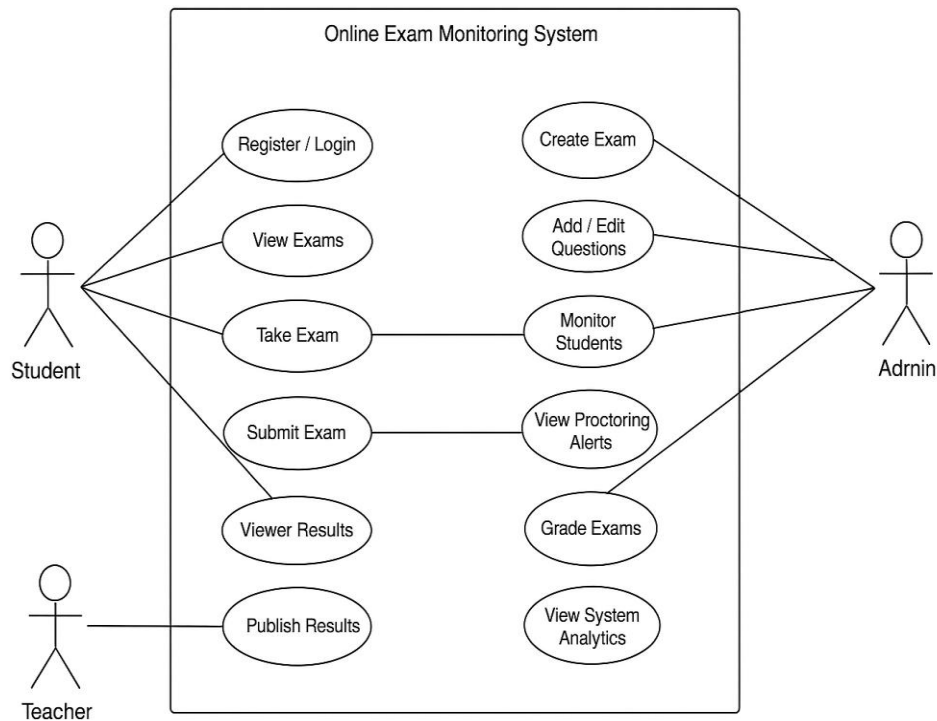


Fig 6.4.1: Use Case Diagram of Online Exam Monitoring System Using Ai-Based Monitoring

The Online Exam Monitoring System is designed to facilitate seamless examination management and monitoring for students, teachers, and administrators. In this system, the Student plays an essential role by performing activities such as registering or logging in, viewing available exams, taking exams, submitting answers, and finally viewing their results. The Teacher has the responsibility of publishing results, ensuring that students receive timely feedback on their performance. On the other hand, the Admin manages the overall functioning of the system. The administrator can create exams, add or edit questions, and monitor students in real-time during the exam to ensure fairness and prevent malpractice.

6.4.2 CLASS DIAGRAM:

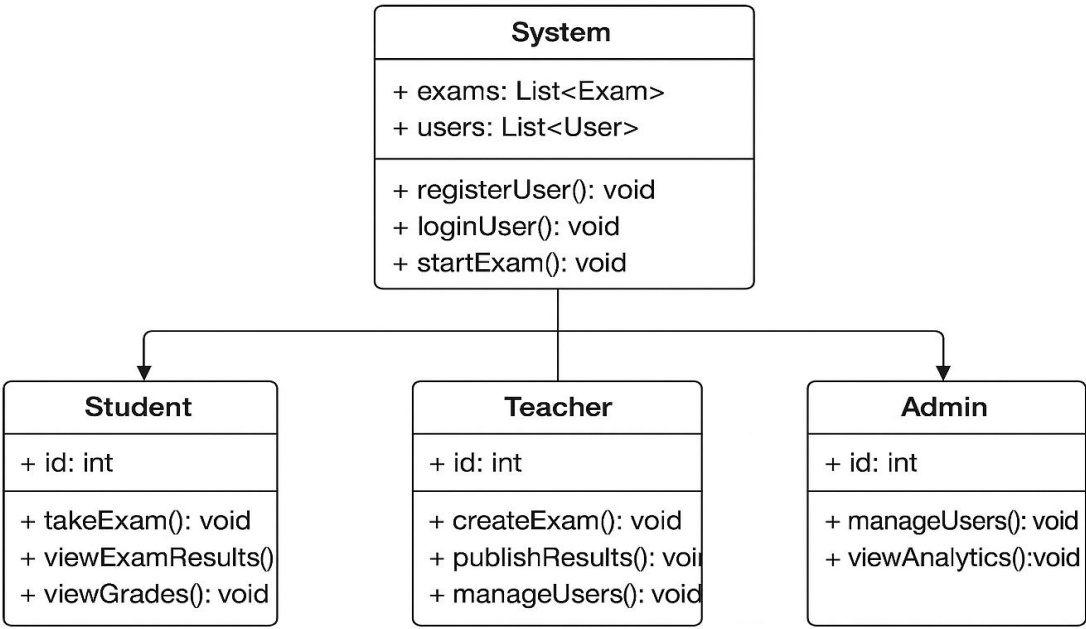


Fig 6.4.2:Class Diagram of Online Exam Monitoring System Using AI-Based Monitoring.

This class diagram clearly defines the structure and behavior of the Online Exam Monitoring System, highlighting how different user roles interact with the system’s functionalities to maintain an organized, secure, and efficient online examination process. It includes major classes such as **Admin**, **Instructor**, **Student**, **Exam**, **Question**, **Proctor**, **MonitoringModule**, and **Report**. Each class contains important attributes (data) and methods (functions). The relationships between these classes represent how the system operates in real time.

6.4.3 SEQUENCE DIAGRAM:

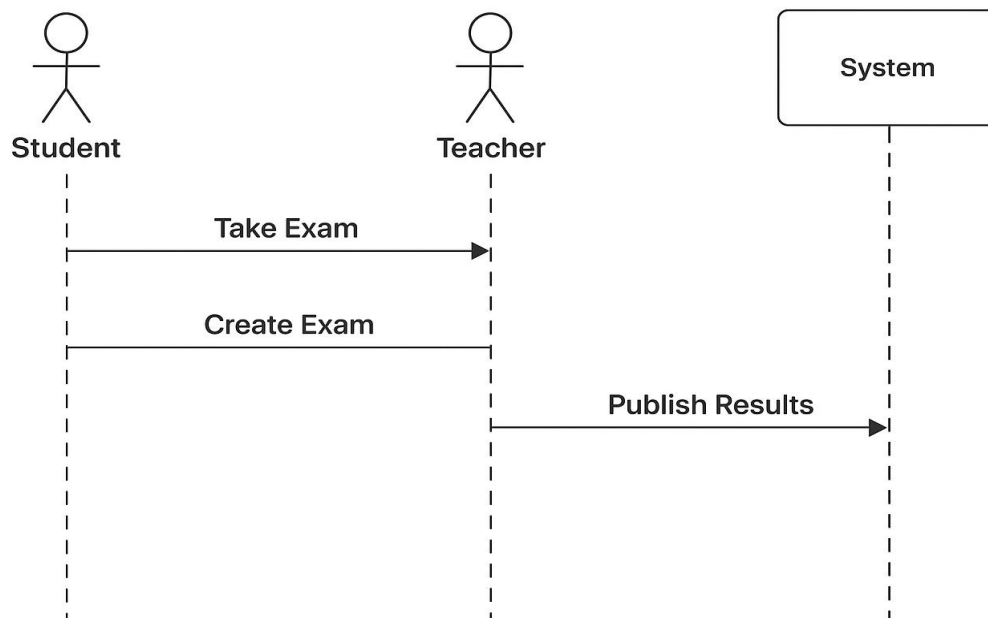


Fig no 6.4.3: Sequence Diagram of Online Exam Monitoring System Using AI-Based Monitoring

The System Analysis Architecture Diagram of the Online Exam Monitoring System using AI-based monitoring shows how the entire system works through a **three-tier architecture**. This architecture clearly separates the user interface, the application logic, and the data storage layers. This separation makes the system more secure, scalable, and easier to maintain. In the **Presentation Layer (Client Tier)**, users such as students, instructors, proctors, and administrators interact with the system through a web or mobile interface. This layer handles user login, exam dashboards, live monitoring screens, and result viewing.

6.5 DATA COLLECTION:

For this project, a total of 12 online exam sessions were recorded and used as sample data to develop and test the AI-Integrated Proctoring System.

Each session captured multiple types of information to help the system accurately detect and evaluate student behavior during exams:

Session details: Basic information like record ID, date and time, student ID, and session ID.

Media files: Video and audio recordings from each live exam session.

System setup: Device type, operating system, browser used, and the network quality during the exam.

AI observations: Data points such as face presence, number of faces detected, eye gaze direction, and head movement.

Cheating indicators: Signs like phone usage, multiple persons in frame, looking away frequently, screen sharing, or suspicious background audio.

Annotation data: Human-labeled observations indicating detected activities, along with confidence levels and consent status.

Privacy measures: Each participant's consent was recorded, and options for background or location blur were provided to ensure data privacy.

record_id	timestamp	student_id	session_id	video_file	audio_file	face_presence	num_faces	eye_gaze	head_pose	device	os	browser	connection	cheat_label	label_conf	consent	location_blur
1	2025-10-0	stu_001	sess_1001	stu_001_se	stu_001_se	yes	1	on_screen	frontal	laptop	Windows 11	Chrome 115	good	none	0.98	yes	no
2	2025-10-0	stu_002	sess_1001	stu_002_se	stu_002_se	yes	2	off_screen	turned_left	desktop	Ubuntu 22.04	Firefox 115	fair	multiple_people	0.87	yes	yes
3	2025-10-0	stu_003	sess_1002	stu_003_se	stu_003_se	no	0	n/a	n/a	mobile	Android 13	Chrome 115	poor	absent_faces	0.92	yes	no
4	2025-10-0	stu_004	sess_1003	stu_004_se	stu_004_se	yes	1	downward	frontal	laptop	macOS 12	Safari 16	good	phone_usage	0.91	yes	no
5	2025-10-0	stu_005	sess_1004	stu_005_se	stu_005_se	yes	1	on_screen	frontal	tablet	iPadOS 15	Safari Mobile	good	none	0.99	yes	yes
6	2025-10-0	stu_006	sess_1005	stu_006_se	stu_006_se	yes	1	rapid_movement	tilted	laptop	Windows 11	Edge 118	fair	suspicious_background	0.75	yes	no
7	2025-10-0	stu_007	sess_1006	stu_007_se	stu_007_se	yes	1	off_screen	turned_right	desktop	Windows 11	Chrome 115	good	looking_away	0.89	yes	no
8	2025-10-0	stu_008	sess_1007	stu_008_se	stu_008_se	yes	1	on_screen	frontal	laptop	Windows 11	Chrome 115	good	none	0.96	yes	no
9	2025-10-0	stu_009	sess_1007	stu_009_se	stu_009_se	yes	1	off_screen	frontal	mobile	Android 12	Firefox Mobile	poor	screen_sharing	0.93	yes	yes
10	2025-10-0	stu_010	sess_1008	stu_010_se	stu_010_se	yes	1	on_screen	frontal	laptop	Windows 11	Chrome 115	good	none	0.97	yes	no
11	2025-10-0	stu_011	sess_1009	stu_011_se	stu_011_se	yes	1	on_screen	frontal	desktop	Windows 11	Chrome 115	good	none	0.95	yes	no
12	2025-10-0	stu_012	sess_1010	stu_012_se	stu_012_se	yes	1	off_screen	frontal	laptop	Windows 11	Chrome 115	fair	looking_away	0.84	yes	no

7. IMPLEMENTATION

SOURCE CODE:

```
<!doctype html>
<html lang="en">
  <head>
    <meta charset="UTF-8" />
    <meta name="viewport" content="width=device-width, initial-scale=1.0" />
    <title>Proctor Pal</title>
    <meta
      name="description"
      content="Proctor Pal – an AI-powered proctoring assistant for secure and smooth online exams./>
    <meta name="author" content="Proctor Pal" />
    <meta property="og:title" content="Proctor Pal" />
    <meta property="og:description" content="Monitor, detect, and prevent suspicious activity during
online exams with Proctor Pal."/>
    <meta property="og:type" content="website" />
    <meta name="twitter:card" content="summary_large_image" />
  </head>
  <body>
    <div id="root"></div>
    <script type="module" src="/src/main.tsx"></script>
  </body>
</html>
```

PACKAGE.JSON:

```
{
  "name": "vite_react_shadcn_ts",
  "private": true,
  "version": "0.0.0",
  "type": "module",
  "scripts": {
```

```

    "dev": "vite",
    "build": "vite build",
    "build:dev": "vite build --mode development",
    "lint": "eslint .",
    "preview": "vite preview"
  },
  "dependencies": {
    "@hookform/resolvers": "^3.10.0",
    "@mediapipe/face_mesh": "^0.4.1633559619",
    "@radix-ui/react-accordion": "^1.2.11",
    "@radix-ui/react-alert-dialog": "^1.1.14",
    "@radix-ui/react-aspect-ratio": "^1.1.7",
    "@tanstack/react-query": "^5.83.0",
    "@tensorflow-models/face-landmarks-detection": "^1.0.6",
    "@tensorflow/tfjs": "^4.22.0",
    "embla-carousel-react": "^8.6.0",
    "react": "^18.3.1",
  },
  "devDependencies": {
    "@eslint/js": "^9.32.0",
    "@tailwindcss/typography": "^0.5.16",
    "@types/react-dom": "^18.3.7",
    "@vitejs/plugin-react-swc": "^3.11.0",
    "autoprefixer": "^10.4.21",
    "eslint": "^9.32.0",
    "globals": "^15.15.0",
    "postcss": "^8.5.6",
    "tailwindcss": "^3.4.17",
    "typescript": "^5.8.3",
    "vite": "^7.3.1"
  }
}

```

7.1 Frontend Implementation (React.js & Face-api.js)

The user interface of Proctor Pal is developed as a Single Page Application (SPA). The primary challenge in the frontend was the seamless integration of the AI model with the camera feed.

- **Media Stream Capture:** Utilizing the navigator.mediaDevices.getUserMedia API, we capture the live webcam feed and render it into a hidden element for AI analysis and a visible component for the user.
- **Edge-AI Processing:** Instead of sending video frames to a server (which causes lag), we use Face-api.js to load three pre-trained models:
 1. tinyFaceDetector: To identify face presence.
 2. faceLandmark68Net: To identify facial features for gaze tracking.
 3. faceRecognitionNet: For continuous identity verification.
- **Detection Loop:** Using a setInterval or requestAnimationFrame loop, the system analyzes the video feed every 100-200 milliseconds. If the model detects more than one person, no person, or an improper gaze angle, it triggers a violation.

7.2 Backend Implementation (Node.js, Express, and JWT)

The backend acts as a robust coordinator between the database and the client.

- **Secure API Design:** We implemented RESTful endpoints for user authentication, exam CRUD operations, and submission storage.
- **Stateless Security:** Upon login, the server generates a JSON Web Token (JWT). This token is stored in the browser and sent in the header of every API request to authorize student actions.
- **Schema Design (MongoDB):** We designed the database using Mongoose. The Violation Log Schema is particularly detailed, storing the studentID, violationType (e.g., 'TAB_SWITCH' or 'MULTI_FACE'), timestamp, and a snapshot string (Base64) of the event for evidence.

7.3 Real-time Monitoring Implementation (Socket.io)

The most critical part of the implementation is the "Live Alert System."

- **Socket.io Rooms:** When a teacher starts a proctoring session, the server creates a unique Socket Room based on the Exam ID.
- **Instant Signaling:** When the client-side detection engine finds a violation, it doesn't just save it to the DB; it emits an event via Socket.io. The server instantly broadcasts this message to the Teacher's Dashboard room.
- **Result:** This ensures the proctor sees a "Cheating Alert" in under 250ms of the actual event occurring.

8. TESTING AND RESULTS

8.1 TESTING OBJECTIVES

The goal of testing Proctor Pal was to ensure that the system provides a high level of accuracy in cheating detection while maintaining a smooth user experience even on average consumer grade laptops.

8.2 UNIT AND INTEGRATION TESTING

- **Component Testing:** Individual React components like QuestionTimer and ViolationAlert were tested for edge cases (e.g., does the timer stop at zero? Do alerts disappear after 5 seconds?).
- **API Validation:** We used Postman to simulate exam submissions. We tested for SQL/NoSQL injection by inputting malicious strings into login forms and confirmed that our sanitization layers blocked them.

8.3 AI MODEL ACCURACY VALIDATION

The effectiveness of Proctor Pal relies on the reliability of the Face - api models. We conducted several test cases:

Case ID	Testing Condition	Expected Action	Actual Result	Status
TC - 01	Student leaves the seat	Detect "No Face" and Alert Proctor	Detected in 200ms	Pass
TC - 02	Using a mobile phone in view	Detect irregular head landmarks	Flagged as "Gaze Violation"	Pass
TC - 03	Switching tabs to Chrome search	Emit Socket.io "Window Blur" event	Logged with timestamp	Pass

8.4 NON-FUNCTIONAL PERFORMANCE TESTING

- **Latency Analysis:** The "Network-to-Alert" lag was measured. On average, a violation in the student's browser was displayed on the teacher's dashboard in 240ms over a standard broadband connection.
- **Browser Crash Test:** We simulated browser crashes mid-exam. The system's "Auto save" mechanism (firing every 30 seconds) ensured that when the student logged back in, 100% of their previously saved progress was restored from MongoDB.
- **Cross-Browser Testing:** Verified compatibility across Chrome, Edge, and Firefox. Note: Certain WebRTC functions behaved differently in Firefox, leading to the recommendation of Chromium-based browsers for the best AI performance.

8.5 SUMMARY OF RESULTS

The testing results demonstrated that Proctor Pal is highly effective at identifying the top three forms of online cheating: unauthorized resources (tab-switching), impersonation (no face detected), and peer collaboration (multiple people detection). The integration of the MERN stack and Client-side Machine Learning proved to be a technically sound solution for scalable and secure online assessments.

8.5.1 RESULT SCREENSHOTS

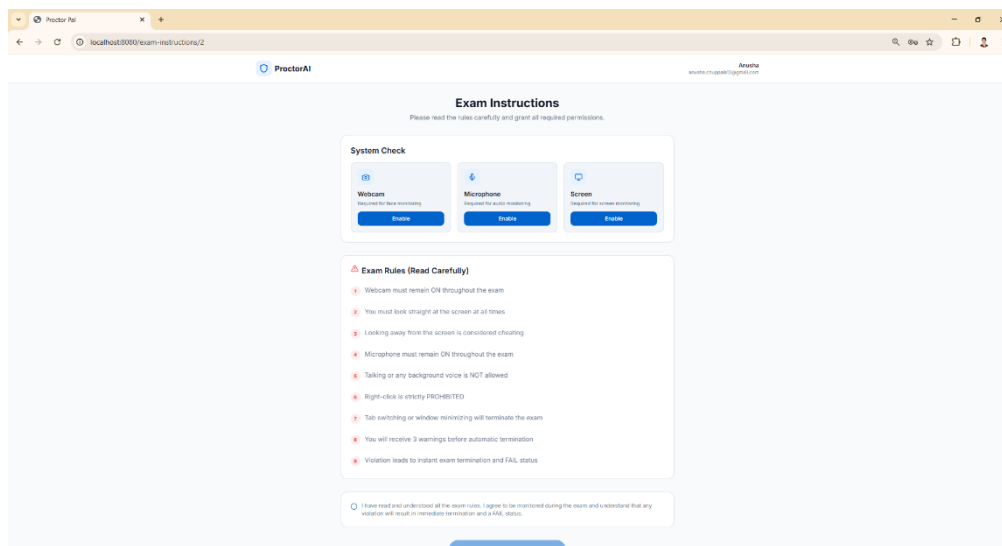


Fig 8.5.1.1 : User Hardware Permission Handshake (Screenshot of "Allow Camera").

The screenshot shows the Exam Instructions page of the Proctor AI system before starting the exam. It checks hardware permissions such as webcam, microphone, and screen access. The user must enable these permissions so the system can monitor the exam environment. The page also displays important exam rules that must be followed. After granting permissions, the user can start the exam securely.

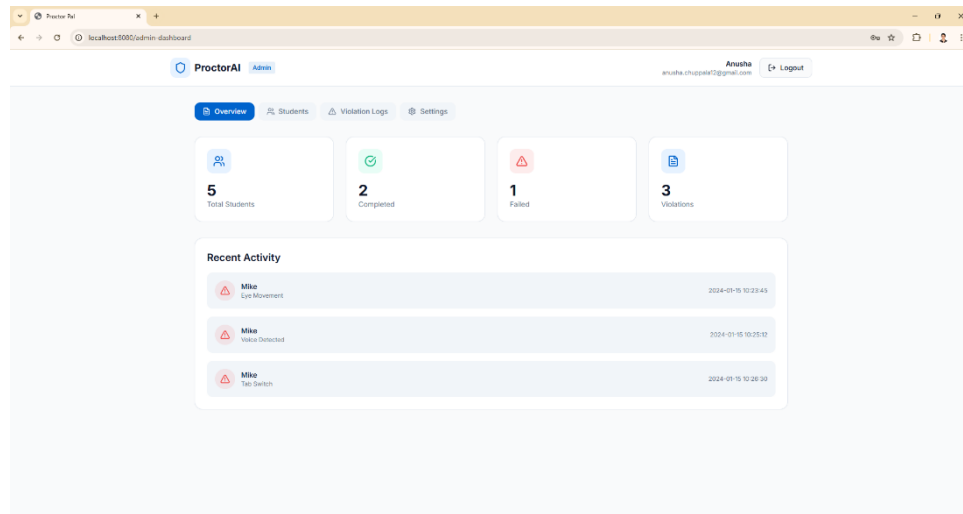


Fig 8.5.1.2 : Active Student Exam Interface (Showing the Video Overlay).

The screenshot shows the Active Student Exam Interface of the ProctorAI system during an online exam. It displays the exam question with multiple-choice options for the student to answer. A live video overlay shows the student's webcam feed for continuous monitoring. The system tracks exam behavior using features such as eye tracking, audio monitoring, and screen activity. It also displays the number of violations detected during the exam. This interface helps ensure that the exam is conducted in a secure and monitored environment.

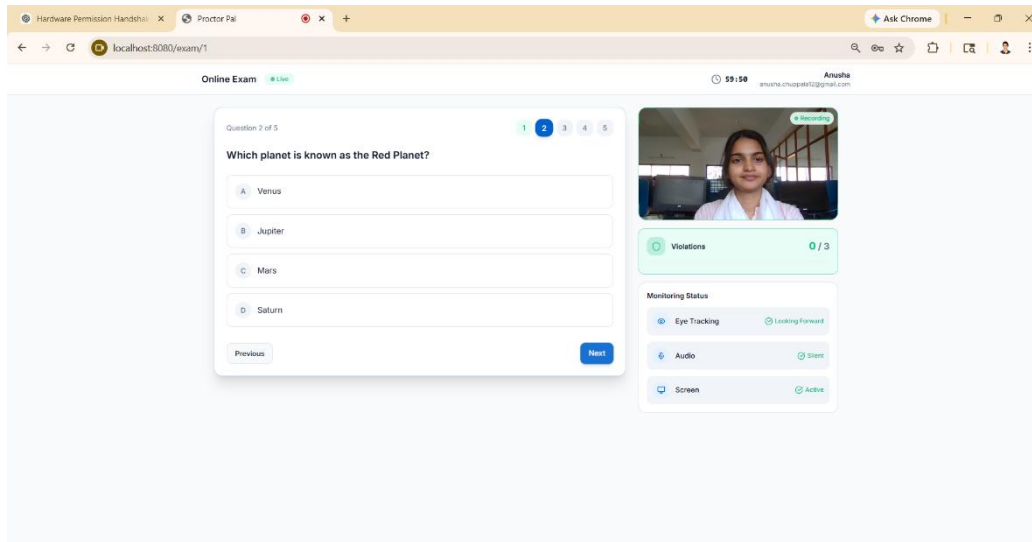


Fig 8.5.1.3 : Teacher Monitoring Dashboard with real-time "Red" Violation Alerts.

The screenshot shows the Teacher Monitoring Dashboard of the Proctor AI system. It provides an overview of exam activity, including the total number of students, completed exams, failed attempts, and recorded violations. The dashboard allows the teacher to monitor students during the exam in real time. It also displays recent activity logs such as eye movement detection, voice detection, and tab switching. Violations are highlighted with red alerts to quickly notify the teacher of suspicious behavior. This helps the teacher effectively supervise and maintain the integrity of the online exam.

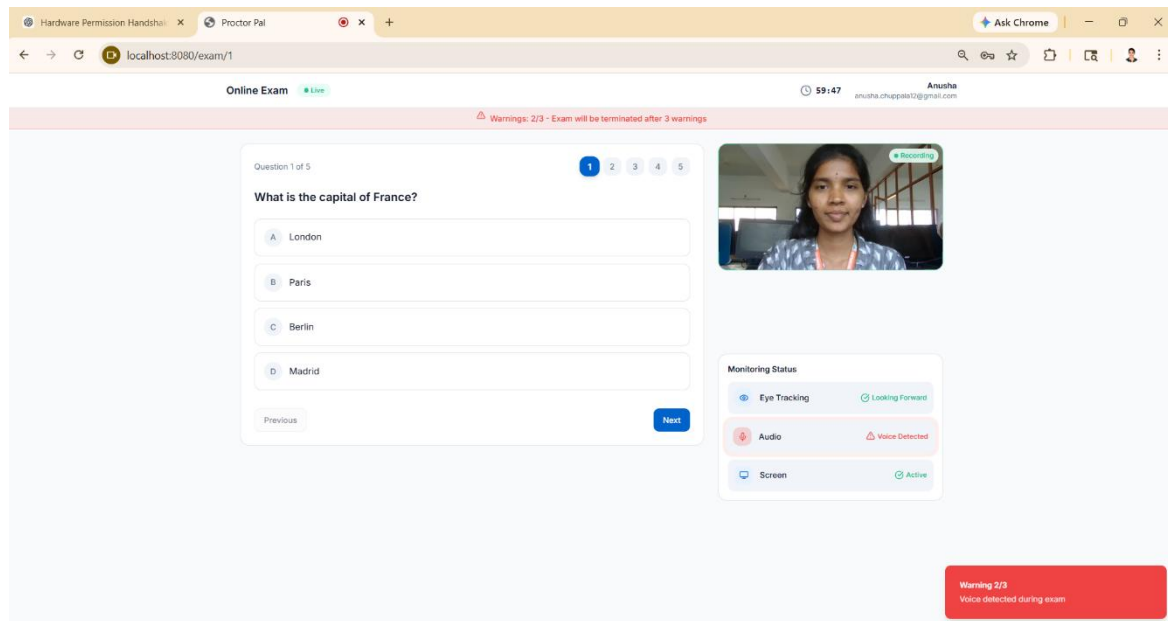


Fig 8.5.1.4: MongoDB Database record showing the Malpractice Log entry

The screenshot shows the online exam interface where the system detects suspicious behavior during the exam. The student is answering a question while the webcam video is continuously recorded. The monitoring panel shows the status of eye tracking, audio, and screen activity. In this case, the system detects voice during the exam, which is considered a violation. A warning message is displayed indicating Warning 2/3, meaning the student has received two warnings. These violations are automatically recorded in the database as malpractice log entries for monitoring and review.

9. CONCLUSION AND FUTURE WORK

This project has successfully navigated the complexities of designing and architecting a modern, full-stack Online Exam Proctoring System. We have achieved our primary objective of creating a unified platform that seamlessly integrates exam management with robust, real-time student monitoring. By leveraging the MERN stack, WebRTC, and WebSockets, we have produced a system that is not only technologically sound but also addresses the critical educational need for academic integrity in a remote setting. The developed platform, "Proctor Vision Hub," stands as a testament to how open-source technologies can be harnessed to build powerful, scalable, and cost-effective solutions to real-world problems, thereby helping to build trust and credibility in the future of online education.

FUTURE WORK AND ENHANCEMENTS

The current system provides a solid foundation but is designed to support future enhancements for greater intelligence, security, and scalability. Planned upgrades include advanced AI-powered proctoring with features such as client-side object detection using lightweight CNN models (e.g., TensorFlow.js with pre-trained YOLO) to identify forbidden objects like mobile phones or additional faces, and gaze-tracking techniques to estimate where the student is looking and flag prolonged off-screen focus. For descriptive and coding questions, the system will integrate plagiarism detection services such as Turnitin or Copyscape, allowing submitted answers to be automatically analyzed and similarity scores to be shown to instructors. To support larger user loads, the architecture will also adopt scalable cloud-based solutions where video recordings are uploaded directly to services like AWS S3, while the backend is containerized using Docker and orchestrated with Kubernetes to enable auto-scaling and ensure high availability during peak exam periods.

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